Multivitamin supplement use and risk of breast cancer: a meta-analysis.

BACKGROUND: The association between consumption of multivitamins and breast cancer is inconsistent in epidemiologic studies. OBJECTIVE: To perform a meta-analysis of cohort and case-control studies to evaluate multivitamin intake and its relationship with breast cancer risk. METHODS: The published literature was systematically searched and reviewed using MEDLINE (1950 through July 2010), EMBASE (1980 through July 2010), and the Cochrane Central Register of Controlled Trials (The Cochrane Library 2010 issue 1). Studies that included specific risk estimates were pooled using a random-effects model. The bias and quality of these studies were assessed with REVMAN statistical software (version 5.0) and the GRADE method of the Cochrane Collaboration. RESULTS: Eight of 27 studies that included 355,080 subjects were available for analysis. The total duration of multivitamin use in these trials ranged from 3 to 10 years. The frequency of current use in these studies ranged from 2 to 6 times/week. In analyses by duration of use 10 years or longer or 3 years or longer and by frequency 7 or more times/week that were reported in these studies, multivitamin use was not significantly associated with the risk of breast cancer. Only 1 recent Swedish cohort study concluded that multivitamin use is associated with an increased risk of breast cancer. The results of a meta-analysis that pooled data from 5 cohort studies and 3 case-control studies indicated that the overall multivariable relative risk and odds ratio were 0.10 (95% CI 0.60 to 1.63; p = 0.98) and 1.00 (95% CI 0.51 to 1.00; p = 1.00), respectively. The association was not statistically significant. CONCLUSIONS: Multivitamin use is likely not associated with a significant increased or decreased risk of breast cancer, but these results highlight the need for more case-control studies or randomized controlled clinical trials to further examine this relationship.